

Fine-Grained Health Monitoring of Optical Modules in Computing Clusters using Transformers

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Abstract—Health monitoring of optical modules in large-scale AI computing clusters faces critical challenges: the extreme scarcity of negative samples hindering model training, and coarse binary ("normal/faulty") classification preventing proactive maintenance by obscuring gradual degradation. To overcome these limitations, this paper presents a fine-grained health assessment framework with Transformer auto-encoders. It incorporates expert-guided anomaly detection, leveraging expert knowledge to annotate initial faults and training the auto-encoder solely on normal data, eliminating reliance on scarce fault samples. The Transformer captures long-term temporal dependencies in multivariate time-series, enabling a 6-level health system (L0-L5) based on reconstructed errors, covering from inactive to irreversible fault for proactive maintenance. To mitigate misjudgment risks, a two-level verification mechanism calibrates samples initially classified into high-severity levels (L4-L5) using physical parameter thresholds. Experiments on 16,335,000 samples (0.00015% faults) validate its effectiveness in identifying degradation, supporting proactive maintenance.

Keywords—Optical modules, Health Assessment, Transformer, Computing clusters

I. INTRODUCTION

Recently, the development of AI models has continuously increased computing power demands. The cluster scale of AI models has grown to tens or even hundreds of thousands of accelerator cards. This expansion may cause more frequent network faults and training interruption [1]. The challenge is concretely illustrated by Meta's Llama 3.1 training run: during 54 days of operation on a 16,384-node H100 GPU cluster, the system experienced 466 interruptions, with 8.4% traced to network failures (including optical modules) [2]. Such disruptions directly compromise the continuity of large-scale AI development.

Therefore, the health state of optical modules is directly related to the stability of the entire cluster network. The fault of optical modules includes launch power degradation, bias current anomaly, temperature rise, laser anomaly, wavelength drift, signal performance imperfection, and receiving sensitivity deterioration etc. [3]. Minor faults may cause decreased data transmission rates and increased error rates, while serious faults could lead to communication interruption. Furthermore, as packaged active components, optical modules present particularly challenging fault management requirements [4].

Over the past few years, machine learning has been widely used in the field of failure management in optical network [5], [6], [7], [8], [9], [10]. Transformer models utilize positional encoding to enable single-pass inference over full input sequences, significantly accelerating detection speed to meet the low latency requirements of fault detection [11], [12]. Meanwhile, Transformers' robust feature extraction capabilities with complex data and effective capture of long-range dependencies demonstrate considerable strengths across time-series task [13]. In the domain of health assessment of optical modules, existing classification methodologies exhibit significant limitations in practical applications. Early studies [14] classified the status of optical modules into two categories: normal and faulty for fault prediction. While operationally straightforward, merely distinguishing between the "normal" and "faulty" states of optical modules often fails to capture early-stage deterioration or subtle anomalies. Other studies [15] introduced enhanced granularity through a six-state classification system: faults, risks, abnormal data, insufficient data, not-started and normal. Although this multi-categorical framework advanced, its models trained on historical fault data suffer from performance limitations when negative samples are insufficient. Furthermore, its threshold-based state classification using Digital Diagnostic Monitoring (DDM) struggles to distinguish subtle degradation differences due to the coarse-grained nature.

In addition, most optical modules in data centers remain healthy for a long time, so the limited negative samples make it challenging for data-driven technology to establish accurate decision boundaries [16]. Concurrently, this imbalance renders AI models overly biased toward healthy samples while reducing fault detection sensitivity [17].

These limitations delay proactive interventions, allowing minor issues to escalate into severe faults that require longer recovery times and more expensive repairs. In contrast, fine-grained health assessment can identify nuanced changes in module status, enabling operators to schedule maintenance during non-critical periods, reduce unplanned downtime, and optimize resource allocation for repairs—ultimately lowering overall maintenance costs while enhancing cluster stability. To address these challenges, we propose a fine-grained health assessment framework for optical modules based on Transformer auto-encoders. The main contributions of our study are as follows:

- Expert-guided anomaly detection: Leveraging expert domain knowledge to annotate fault samples, the auto-encoder is trained exclusively on normal operating data. This eliminates the dependency on fault samples during model training while ensuring sensitivity to anomalous patterns.
- Temporal modeling with hierarchical assessment: The Transformer architecture captures long-term temporal dependencies in multivariate time-series data. We establish a 6-level health assessment system (L0–L5) based on the distribution of reconstructed errors. This hierarchy comprehensively covers the operational life cycle of optical modules— (inactive, optimal, stable , sub-health, minor fault and critical fault) —enabling operators to make precise maintenance scheduling decisions.
- Two-level verification mechanism: Following the initial classification of health grades (L1-L5), samples initially categorized as higher grades (L4-L5) undergo downgrading and calibration through integration of physical-parameter thresholds. This mechanism serves to mitigate the high risks that may arise from model misjudgments.

The rest of this paper is organized as follows. Section II describes the Transformer-based fine-grained health assessment framework and its operational workflow. Section III details the characteristics of the datasets, experimental setup, evaluation metrics and results. Finally, Section IV concludes this study.

II. TRANSFORMER-BASED FINE-GRAINED HEALTH ASSESSMENT METHOD

A. Data Selection and processing

The historical data of the optical modules encompasses DDM parameters and error packet counts retrieved from device ports. The DDM parameters include transmitted optical power, received optical power, bias, voltage and temperature.

Linear interpolation is used to fill the missing values in the dataset. To ensure that all features are on a comparable scale and to mitigate the impact of dimensional differences, the z-score normalization method is applied. The specific equation is as follows:

$$z = \frac{(x - \mu)}{\sigma} \quad (1)$$

where z is the normalized value of the future, x is the original value of the future, μ is the mean, and σ is the standard deviation.

B. Health assessment framework

The Transformer-based fine-grained health assessment framework is illustrated in Fig. 1:

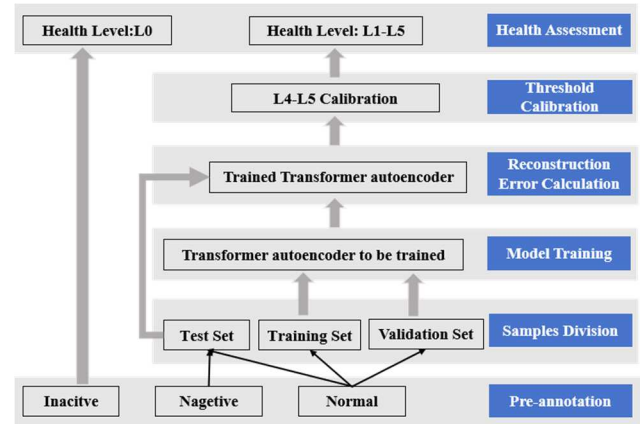


Figure 1. Health assessment framework

In the data processing stage, samples are categorized into three distinct groups: inactive, negative, and normal. Inactive samples are directly labeled as L0 and excluded from optical module health degree calculation. The normal samples are partitioned into a training set, a validation set and a partial test set. Subsequently, the partial test set is merged with the faulty samples to form the final test set. The training set serves as the foundation for training a health assessment model based on the Transformer architecture. The trained health assessment model is capable of learning the probability distribution of normal state samples, enabling it to accurately reconstruct normal state samples, while failing to correctly reconstruct abnormal state samples. Consequently, the reconstruction error of the Transformer model can effectively quantify the divergence between abnormal and normal samples learned by the model. The health status of optical modules is characterized by the degree of deviation of test samples from the distribution of normal state samples. To re-calibrate high-grade optical modules (L4-L5), we set thresholds for each parameter and downgrade the samples to L3 if any parameter exceeds the threshold.

The Transformer auto-encoder employs a symmetric encoder-decoder structure to learn normal operational patterns of optical modules. By integrating positional encoding and multi-head self-attention mechanisms, this architecture enables quantification of health degradation through reconstruction error analysis. The model's technical components are structured as follows:

- **Input Embedding Layer:** Raw input features (6 dimensions) are projected into a 16-dimensional latent space ($d_{model}=16$) via a linear transformation. A ReLU activation is applied, followed by dropout regularization ($p=0.3$) to mitigate overfitting. A layer normalization operation is applied to stabilize the training process by normalizing the feature distribution across the latent space.
- **Positional Encoding Layer:** Sine-cosine positional encoding injects sequence order information using the standard formulation.
- **Encoder Module:** Composed of 2 stacked Transformer encoder layer, which integrates a multi-head attention mechanism ($n-head=2$) and a feed-forward neural

network with a hidden dimension, followed by layer normalization and residual connections to stabilize training.

- **Decoder Module:** Designed symmetrically with 2 Transformer decoder layer, which utilizes a masked multi-head self-attention mechanism to process the encoded sequence and a cross-attention mechanism that uses the encoder outputs as memory units to guide the reconstruction process.
- **Output Projection:** The 16-dimensional decoder output is projected back to the original 6-dimensional feature space via linear transformation, enabling the calculation of Mean Squared Error (MSE) as the reconstruction loss.

The discrepancy between the input test sample and its average reconstructed sample is defined as the reconstruction error:

$$RE(x_i) = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2 \quad (2)$$

where n is the feature dimension, x_i is the i -th dimension feature value of the normal feature vector and $\hat{x}_i$ is the i -th dimension feature value of the reconstructed feature vector.

The health score mapping framework is established by analyzing the reconstruction error distribution derived from the training set:

$$Health\ Score = \begin{cases} 100, & TRE < E_{min} \\ 100 - 30 \cdot \frac{TRE - E_{min}}{\Delta E}, & E_{min} \leq TRE \leq E_{max} \\ 70 - 20 \cdot \frac{TRE - E_{max}}{0.5\beta \cdot \Delta E}, & E_{max} < TRE \leq MFRE \\ 50 \cdot \left(1 - \min\left(1, \frac{TRE - CFRE}{0.5CFRE}\right)\right), & TRE > MFRE \end{cases} \quad (3)$$

where E_{min} and E_{max} represent the minimum and maximum reconstruction error of the training set respectively, TRE denotes the reconstruction error of test samples, β is the sensitivity coefficient (2-5), which governs sensitivity to fault severity, and ΔE is the difference between E_{min} and E_{max} . $CFRE$ is the upper limit of the reconstruction error for critical fault samples:

$$CFRE = E_{max} + \beta \Delta E \quad (4)$$

$MFRE$ is the upper limit of the reconstruction error for minor fault samples:

$$MFRE = E_{max} + 0.5\beta \Delta E \quad (5)$$

Based on the health score, the health status of optical modules is categorized into six levels (L5 to L0), with a comprehensive evaluation provided in Table I, which establishes a complete health management system for optical modules while offering corresponding operation and maintenance recommendations.

TABLE I. HEALTH ASSESSMENT STANDARD

Level	Health Score	Status Definition	Characteristics	Maintenance Recommendations
5	95-100	Optimal State	All parameters within optimal ranges	Regular inspection
4	85-95	Stable State	Occasional minor parameter drift without performance impact	Regular inspection

3	70-84	Sub-health State	Parameter drift approaching fault thresholds	Enhance monitoring and troubleshoot problems
2	50-69	Minor Fault State	Parameters with slight deviation and minor impact on network	Inspect and replace within 12 hours
1	0-49	Critical Fault State	Parameters with severe deviation and serious impact on network	Replace the failure optical modules immediately
0	---	Inactive state	The optical module in a disabled state	---

III. EXPERIMENTAL ANALYSIS

A. Datasets

The datasets are derived from the optical module monitoring data of GPU servers and switches in the intelligent computing cluster of Zhejiang Lab. Data were collected at half-hour intervals over a period of approximately six months. The DDM distribution of optical modules are illustrated in Fig. 2. After data cleaning, a total of 16,335,242 valid samples are retained. Among these, 826,127 optical module samples (5.06% of the total) with a received optical power of -35 dBm are in a disabled state.

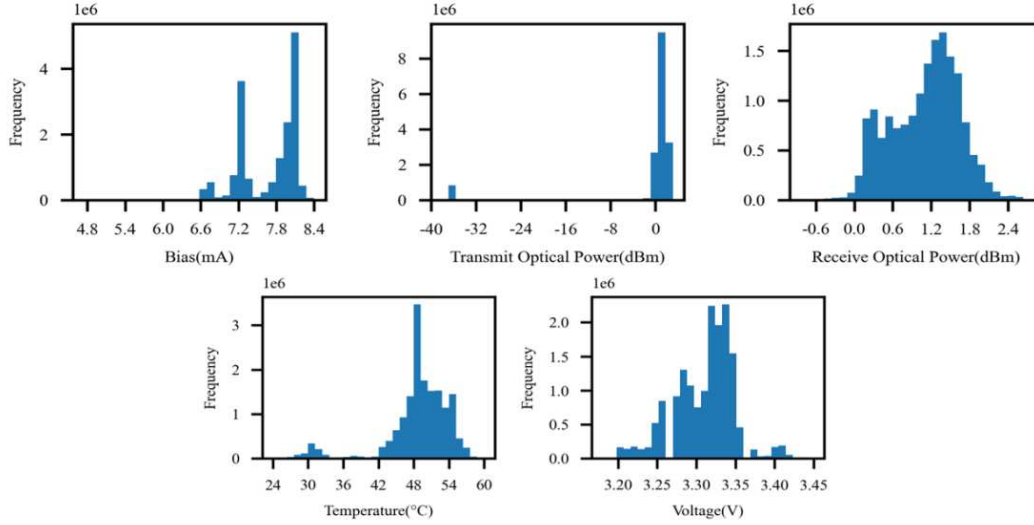


Figure 2. DDM distribution of optical modules

Only five samples exhibited error packet occurrences. Based on expert insights, the operational temperature of an optical module typically exceeds the ambient temperature by 5 to 25 °C. The highest recorded temperature of optical modules in the dataset is 60 °C, implying an ambient temperature of 35 °C. However, such elevated ambient temperature is highly improbable according to the environmental control standards of the computer room. Consequently, 16 samples with temperature above 60 °C are flagged as temperature-anomaly cases. In total, there are 21 initial negative samples, accounting for only 0.00015% of the entire dataset.

B. Experimental Settings

After excluding inactive samples and initial faulty samples, the remaining samples underwent a two-stage split. First, 20% were allocated to the test set, while the remaining 80% formed the combined training and validation set pool. This pool was then subdivided at a 4:1 ratio into a training set (64% of the original remaining samples) and a validation set (16% of the original remaining samples). This resulted in a final distribution of normal samples as follows: 60% training, 20% validation and 20% test.

To comprehensively evaluate model performance, a consolidated test set was constructed. This set encompassed three operating conditions: the 20% normal samples held out initially, 21 initial negative samples, and 826,127 inactive samples, yielding a total of 3,927,967 samples. The final training and validation sets contained 9,305,456 and 3,101,819 samples, respectively.

C. Evaluation Metrics

To quantify Transformer model performance, we utilize raw framework outputs (i.e., initial health grades before applying downgrading L4-L5 samples). This analysis demonstrates the necessity of two-level verification by revealing potential misjudgments in high-severity samples (L4-L5).

Due to the limited negative samples, the imbalance may lead to a tendency to classify samples as normal, which deteriorates the assessment performance for negative samples. In such scenarios, while accuracy may remain relatively high, this metric fails to faithfully reflect the model's actual fault assessment capability. To address this, this study designates F1 score as the primary evaluation metric. As the harmonic mean of precision and recall, F1 score integrates the model's classification ability for both negative and normal samples, providing a more reasonable performance assessment under limited fault samples [17].

$$Precision (\%) = \frac{TP}{TP+FP} \times 100\% \quad (6)$$

$$Recall (\%) = \frac{TP}{TP+FN} \times 100\% \quad (7)$$

$$F_1 \text{ score } (\%) = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (8)$$

where TP is the number of true positive classifications, FP is the number of false positive classifications, and FN is the number of false negative classifications.

D. Experimental Results

Prior to physical threshold calibration, 20 abnormal samples are misclassified as high-severity levels (L4-L5). F1 score is about 99.99967%.

After the health assessment of the test set and inactive samples, the distribution of health level is shown in Fig. 3. 826,127 samples are classified as L0. Only one sample is classified as L1 with no L2 samples. Meanwhile, 383 samples are labeled as L3, 311 samples are classified as L4 and a significant 3,101,145 samples are grouped into L5. This stratification of initial normal samples into L2-L4 levels identifies potential threats, providing maintenance technicians with enhanced diagnostic evidence."

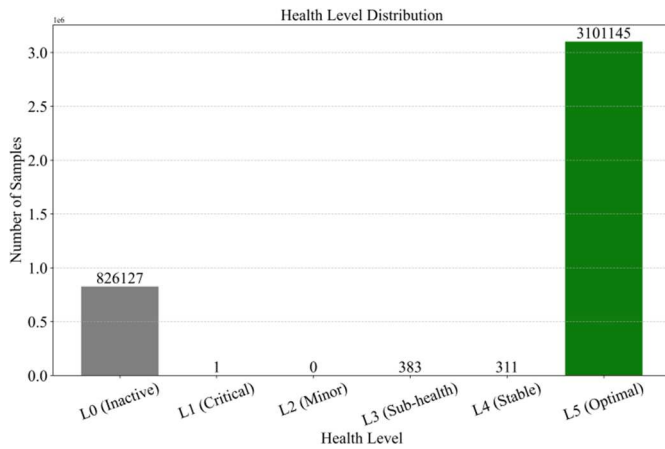


Figure 3. Health level distribution of test set

As shown in Fig. 4, the health scores of L3 samples are predominantly concentrated within the 73-74 range. Specifically, 45.2% of samples score 73 and 42.3% score 74. Minor proportions score 72 (4.2%) and 75 (5.7%). Additionally, a small fraction (0.8% each) scores 98 or 99. These 98-99 scored samples represent cases downgraded from the L5 through the two-level verification mechanism. This mechanism downgrades and calibrates samples initially classified as higher grades (L5) by incorporating equipment physical parameter thresholds, mitigating risks from potential model misjudgments. Notably, all 16 samples scoring 99 exhibit temperature of 60 °C and the 4 samples scoring 98 each show one ingress error packet count. Although these anomalies do not impair network traffic, continuous close monitoring is nonetheless warranted.

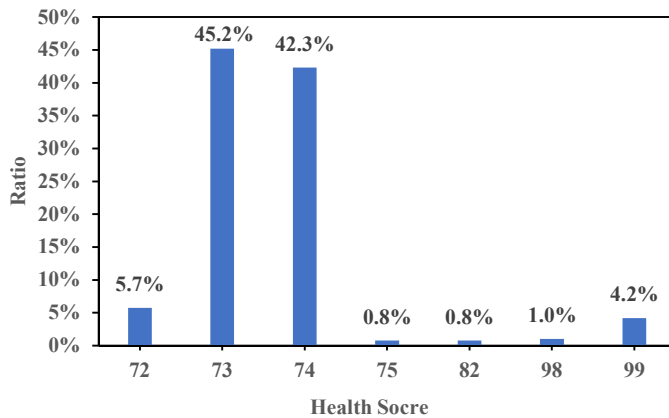


Figure 4. Health score distribution of L3

The test set contain the initial negative samples. As illustrated in Fig. 3, the health levels of initial negative samples distribute in L1 and L2. One sample is classified as L1 and 20 samples are categorized as L3, while no samples are assigned to L2. Analysis the parameters of these samples reveals that the L1 sample exhibited 72 ingress packet errors. For the L3, 16 samples with abnormal temperature of 60 °C and 4 samples with 1 ingress packet error each are downgraded from L5 by the two-level verification mechanism.

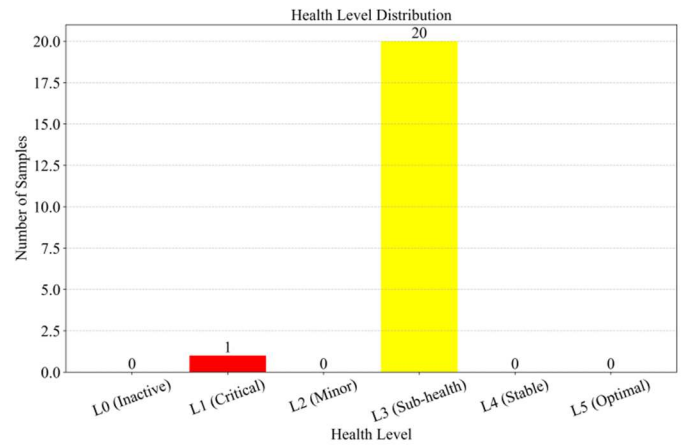


Figure 5. Health level distribution of initial negative samples

In addition to the 20 initial negative samples mentioned above, there are an additional 363 sub-healthy samples. These samples originated from two physical optical modules with model types QSFP-DD and QSFP-28 respectively. Among these, 360 samples are from QSFP-28 modules and 3 from QSFP-DD optical modules. The QSFP-DD samples are classified as L3 due to transient drops exceeding 6 dBm in received optical power followed by recovery to baseline, as shown in Fig. 6. The QSFP-28 samples are classified as L3 because their received optical power (around -6 dBm) is significantly lower than that of other same type optical modules, as illustrated in Fig. 7.

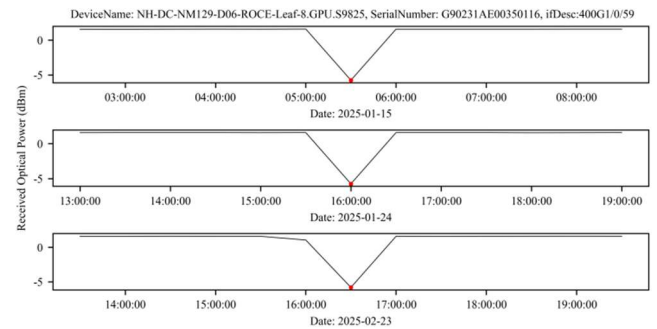


Figure 6. Received optical power timing diagram of QSFP-DD

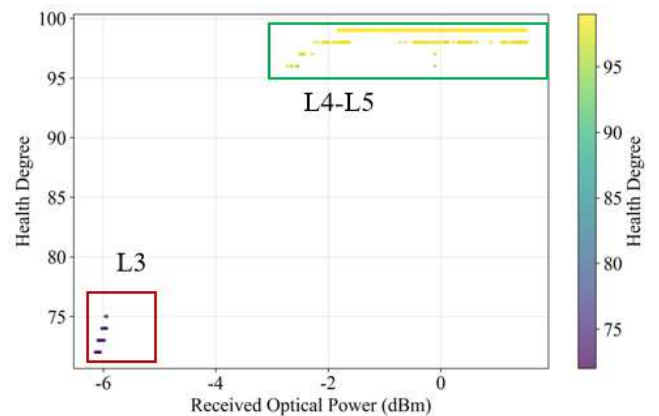


Figure 7. Correlation between received optical power and health degree of all QSPF_28 optical modules.

IV. CONCLUSION

This study introduces a Transformer-based framework for fine-grained health monitoring of optical modules in computing clusters. Experimental results show that the health assessment framework can effectively identify sub-health (L3) states and reduce reliance on negative samples that traditional methods [14], [15] overlook. The six-level health assessment system (L0 – L5) characterizes optical module states from inactive (L0) to optimal (L5), enabling proactive data-driven maintenance. By integrating expert-guided anomaly detection, hierarchical temporal modeling, and a two-level verification mechanism, the proposed framework achieves accurate and rational health assessment under extreme fault sample scarcity (merely 0.00015% in the dataset). The approach not only improves early-risk detection but also reduces dependency on rare fault data, demonstrating high applicability in real-world data centers.

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